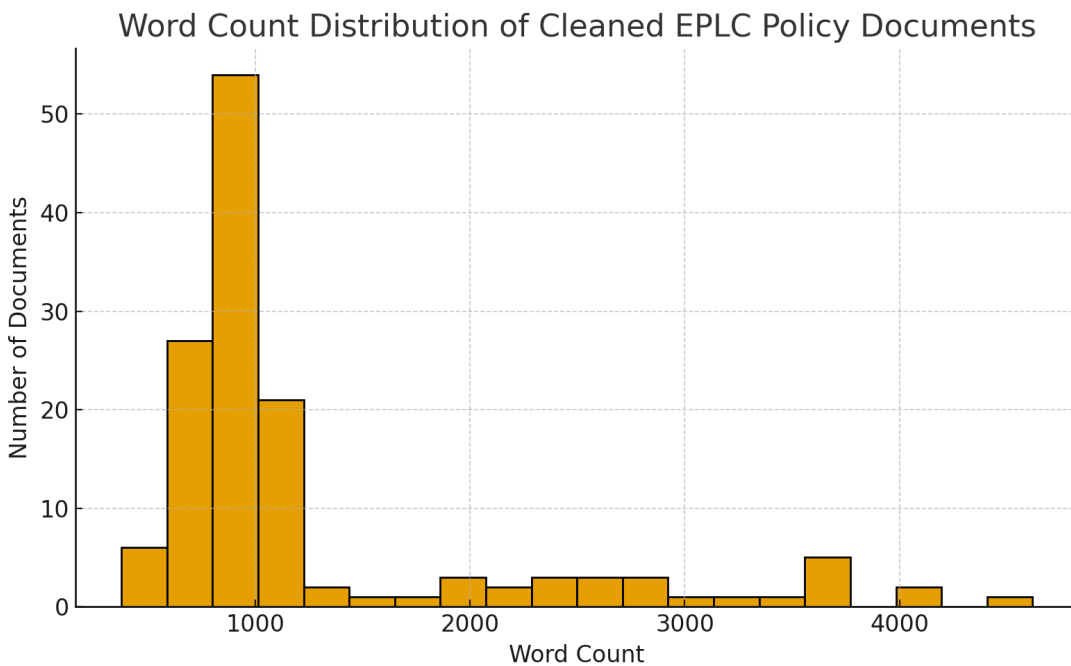


Data Visualization Assignment

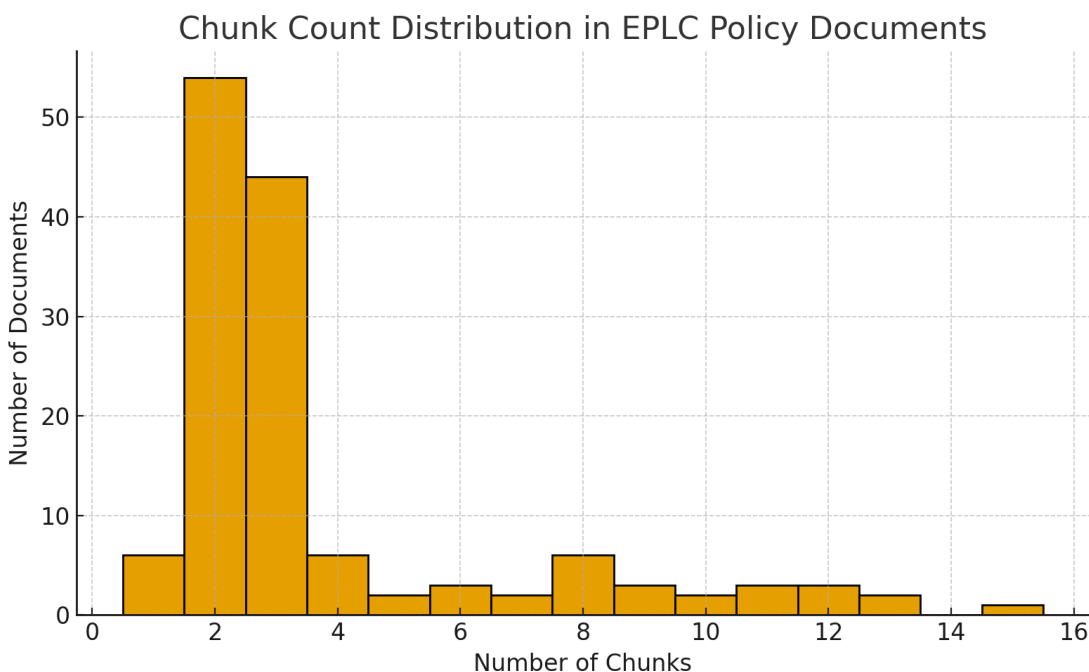
Visualization 1: Word Count Distribution of Cleaned EPLC Policy Documents

This histogram displays the distribution of word counts across the cleaned EPLC policy documents used in our IBM Practicum RAG pipeline. Each document was converted into standardized .txt format during the data-cleaning stage, after which the total word count was computed. This visualization helps us understand the structural characteristics of the dataset and informs decisions about chunking and embedding.



Visualization 2: Chunk Count Distribution After Splitting Documents

This histogram shows how many chunks were generated per document after applying an estimated chunking strategy (approximately 300 words per chunk). The distribution helps predict embedding volume, retrieval complexity, and memory usage in the downstream RAG system.



Document length is highly variable and right-skewed. The majority of EPLC documents fall between 600–1,000 words, while a smaller set extends to 3,000–4,500 words. This long right tail represents complex procedural documents that require much more processing. Understanding this variation is essential for designing a scalable data pipeline.

Chunk counts reflect the structural complexity of the corpus. The chunk distribution shows that most documents generate 2–4 chunks, while a small subset produces 10+ chunks. This directly impacts embedding volume, retrieval granularity latency and cost in inference. Documents with many chunks pose a higher challenge for coherent retrieval.

Implications for embedding strategy. Because only around 20% of documents are long and produce many chunks, we can optimize the pipeline without increasing cost across the entire dataset. For example, adaptive chunk sizes for longer documents, additional metadata (section headers, policy categories), and prioritizing long documents during evaluation. This ensures high retrieval accuracy while keeping embedding costs efficient.

Impact on RAG performance and evaluation. Longer documents, if chunked poorly, can lead to context splitting and retrieval noise. These visualizations highlight the need for semantic-aware chunking (instead of fixed-length splitting), better indexing of long procedural files, and testing retrieval recall separately for short vs. long documents. This will improve the model's ability to produce grounded and policy-specific answers.

In conclusion, both visualizations reveal important structural patterns in the EPLC corpus that directly influence our RAG design choices. Document length, chunk volume, and distribution

characteristics guide decisions about chunking, embedding optimization, and retrieval quality. These insights strengthen our pipeline and help ensure more accurate, efficient, and policy-aligned responses in the IBM Practicum project.